

EVALUATION OF A SMART MOBILE ROBOTIC SYSTEM FOR INDUSTRIAL PLANT INSPECTION AND SUPERVISION

A PROJECT REPORT

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BONAFIDE CERTIFICATE

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ABSTRACT

The increasing complexity and scale of modern industrial plants have necessitated the deployment of advanced automation solutions to enhance operational efficiency, safety, and reliability. This project evaluates a smart mobile robotic system specifically designed for the autonomous inspection and supervision of high-risk industrial environments. To address the limitations of manual monitoring, the proposed robotic system is integrated with a suite of state-of-the-art sensors, including specialized gas sensors for leak detection and flame sensors for early fire identification. These instruments, combined with advanced localization modules, facilitate accurate navigation and comprehensive environment mapping within dense industrial layouts.

The evaluation framework focuses on critical Key Performance Indicators (KPIs) to validate the system's field readiness, specifically measuring navigation accuracy, anomaly detection rates, real-time response latency, and overall system reliability under varying operational conditions. Through rigorous testing, the results demonstrate that the smart mobile robotic system effectively reduces the necessity for human intervention in hazardous zones, thereby minimizing exposure to potential risks. Furthermore, the automation of routine checks significantly reduces total inspection time and enhances the precision of supervision tasks compared to traditional methods.

By providing consistent, high-fidelity data streams and rapid alerts, the system contributes to improved workplace safety and a measurable reduction in unplanned downtime. Ultimately, this research underscores the transformative potential of mobile robotics in transitioning industrial maintenance from reactive to proactive paradigms, ensuring robust protection for both personnel and infrastructure.

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LIST OF ABBREVIATIONS

ACRONY

GSM	GLOBAL SYSTEM FOR MOBILE COMMUNICATIONS
GPS	GLOBAL POSITIONING SYSTEM
IoT	INTERNET OF THINGS
PHP	PERSONAL HOME PAGE
WiFi	WIRELESS FIDELITY
PC	PERSONAL COMPUTER
IDE	INTEGRATED DEVELOPMENT ENVIRONMENT
AVR	AUTOMATIC VOLTAGE REGULATOR

CHAPTER-1

INTRODUCTION

The increasing complexity and scale of modern industrial plants have necessitated the deployment of advanced automation solutions to enhance operational efficiency, safety, and reliability. This project evaluates a smart mobile robotic system specifically designed for the autonomous inspection and supervision of high-risk industrial environments. To address the limitations of manual monitoring, the proposed robotic system is integrated with a suite of state-of-the-art sensors, including specialized gas sensors for leak detection and flame sensors for early fire identification. These instruments, combined with advanced localization modules, facilitate accurate navigation and comprehensive environment mapping within dense industrial layouts.

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By providing consistent, high-fidelity data streams and rapid alerts, the system contributes to improved workplace safety and a measurable reduction in unplanned downtime. Ultimately, this research underscores the transformative potential of mobile robotics in transitioning industrial maintenance from reactive to proactive paradigms, ensuring robust protection for both personnel and infrastructure.

1.1 BACKGROUND

The development of this project is rooted in the evolution of industrial safety and monitoring technologies over the past few decades. In the 1990s, the introduction of fixed sensor networks revolutionized plant maintenance by allowing for basic fault detection through wired alarm systems, marking the beginning of automated facility health monitoring. The 2000s saw the rise of supervisory control and data acquisition (SCADA) systems, primarily used for centralized management, which enabled remote monitoring but lacked mobile flexibility or advanced features like autonomous environment analysis. By the 2010s, the Internet of Things (IoT) began transforming the industrial sector, leading to the emergence of connected factories capable of sharing data across devices, although such systems were mostly limited to stationary machinery. In 2014, the adoption of collaborative robotics brought features like obstacle avoidance and path planning, improving shop-floor safety but still without comprehensive real-time mobile supervision. With the growth of cloud computing by 2017, industrial data could now be stored, accessed, and analyzed remotely, opening the door for predictive maintenance and deep behavioral insights into plant operations. The COVID-19 pandemic in 2020 accelerated the need for remote and automated systems, encouraging a shift toward safer, contactless technologies that reduce human density in plants. By 2022, the fusion of AI and mobile robotics led to intelligent systems that could not only detect but also predict potential hazards, offering preemptive solutions to prevent industrial disasters. Recognizing the limitations of fragmented solutions and the need for a unified, cost-effective, and real-time mobile system, this project was conceptualized in 2024. Its objective was to create a smart monitoring solution that integrates emergency alerts, dynamic robotic response, and centralized data analysis. By 2025, the project culminated in a robust robotic prototype capable of continuously monitoring plant environments, sending instant alerts, adapting to obstacles, and contributing to a safer and more intelligent industrial infrastructure.

1.2 OBJECTIVE

Ensuring industrial safety and environmental monitoring is crucial in modern manufacturing and processing systems. This project aims to enhance plant safety by continuously assessing the atmospheric conditions and overall structural status. By implementing intelligent robotic monitoring, it identifies potential risks such as gas leaks, fire hazards, or sudden changes in environmental parameters. The system actively tracks these conditions in real-time, providing instant feedback and preventive measures to improve facility safety. Additionally, the project focuses on anomaly detection and response mechanisms. It monitors unusual gas concentrations and thermal signatures to determine whether a hazard has occurred. When a threat is detected, the system immediately triggers an alert and shares critical information, allowing for timely assistance. This real-time data transmission helps reduce response time and enhances emergency support for on-site personnel. To further strengthen safety measures, the system integrates a mechanism for adjusting robotic patrol speeds based on environmental data. It enables real-time path variations to ensure that the robot operates within safe limits, minimizing potential collisions. A key aspect of this project is its ability to provide real-time updates and alerts to relevant plant managers and safety officers. The system continuously gathers data and transmits it to a centralized platform for remote monitoring. This ensures that the plant's status, environmental hazards, and robotic performance are promptly recorded, improving response efficiency and overall industrial safety management.

1.3 PROJECT OVERVIEW

This project introduces an intelligent mobile robotic monitoring system that combines sensors, microcontrollers, and wireless communication to collect and analyze real-time industrial data. It monitors various factors such as atmospheric gas levels, thermal anomalies, environmental conditions, and potential structural failures. When a

risk is detected, the system triggers automated alerts and responses, enabling timely action. All the data is transmitted to a centralized cloud-based platform for remote access and analysis. This allows for efficient facility management, predictive maintenance, and enhanced safety for all plant personnel.

1.3.1 ADVANTAGES

Real-time Alerts The system is equipped with sensors and intelligent logic that continuously monitor both environmental hazards and robotic parameters. In case of emergencies—such as gas leaks, fire detection, or collision impact—the system immediately triggers alerts. These alerts are sent instantly to relevant stakeholders, including safety officers, plant managers, or emergency services. This rapid response mechanism ensures timely intervention, which can be critical in preventing serious industrial accidents or minimizing the damage caused by unexpected leaks or fires.

Preventive Safety Beyond responding to incidents, the system proactively enhances safety through early fault detection. By constantly monitoring air quality, temperature gradients, and equipment heat signatures, it can identify issues before they escalate into serious explosions or fires. This preventive approach helps avoid plant shutdowns in critical situations and reduces the likelihood of accidents caused by undetected leaks. It also supports scheduled maintenance of the robotic unit itself, increasing the overall lifespan and reliability of the supervision system.

Data-Driven Insights All data captured by the system—such as gas concentration patterns, temperature maps, and route history—is logged and stored for analysis. Over time, this data builds a comprehensive profile of both the plant's health and the robotic system's performance. These insights can be used to identify high-risk zones, improve ventilation efficiency, schedule maintenance more accurately, and even

train personnel to adopt safer operational practices. In industrial settings, such data can also support safety compliance audits and insurance assessments.

Remote Monitoring One of the key advantages of the system is its ability to operate over the internet, leveraging IoT technology to enable remote access. Authorized users can view live video feeds, receive sensor alerts, and generate safety reports from any location through a centralized web or mobile platform. This is particularly beneficial for safety coordinators and facility managers, allowing them to oversee multiple sectors of a plant in real-time without being physically present in hazardous zones. It streamlines facility supervision and crisis response.

Adaptive Response The system is not just passive in monitoring; it actively adapts to real-time conditions to ensure optimal performance. For example, if high gas levels are detected in one corridor, the robot can prioritize that area for closer inspection; if an obstacle is sensed ahead, it autonomously calculates a new path. By analyzing sensor input and environmental data, the system helps the robot operate within safe and efficient parameters. This dynamic adjustment improves navigation stability and reduces the risk of human error during manual inspections.

1.4 MOTIVATION

The motivation behind this project stems from the alarming increase in industrial accidents due to gas leaks, undetected fires, and delayed inspection routines. Despite significant advancements in industrial technology, many incidents still occur due to the lack of real-time mobile monitoring and immediate response systems. Statistics reveal that a large percentage of plant disasters could be prevented if timely alerts and automated preventive actions were available. By integrating mobile robotics and IoT into industrial plants, there is a significant opportunity to reduce such incidents and enhance workplace safety. The rising demand for safer, more intelligent industrial

environments also reflects a societal shift toward technology-driven worker welfare. Additionally, the absence of affordable, unified mobile monitoring systems for small-to-medium enterprises—especially in developing regions—further emphasized the need for this solution. The desire to contribute to public safety, improve industrial reliability, support smarter factory infrastructure, and leverage modern technology for tangible societal benefit serves as the core inspiration for this work.

1.5 INNOVATION AND NOVELTY

What sets this project apart is its holistic and integrated approach to industrial safety and mobile intelligence. Unlike conventional systems that offer isolated features—such as fixed smoke detectors or manual inspection carts—this project combines multiple advanced capabilities into a unified mobile framework. The system continuously monitors both environmental hazards and its own navigation health, allowing for real-time detection of anomalies such as gas leaks, flame presence, or mechanical faults. What makes it truly innovative is the incorporation of intelligent alert mechanisms that not only detect emergencies but also automatically notify authorities, enabling faster response and minimizing potential consequences. In addition, the system is designed to adapt dynamically to internal conditions—such as changing plant layouts or moving machinery—by initiating safer navigation paths, which contributes to improved efficiency and reduced risk. Cloud integration is another key differentiator, offering centralized data storage, historical trend analysis, and remote access to safety insights, which can be valuable for both safety auditors and plant management. This forward-looking design leverages IoT and real-time analytics, creating a robust platform that goes beyond simple detection and into intelligent autonomous supervision. The novelty lies in its ability to combine affordability, mobility, and next-generation sensing in a practical solution aimed at making industrial environments safer and more responsive.

1.6 SCOPE

This project offers a comprehensive industrial monitoring solution using mobile robotics and IoT technology to ensure facility safety. It provides real-time data acquisition from various sensors such as gas, flame, and ultrasonic detectors to monitor both the environment and the robot's navigation status. The system can detect emergencies like hazardous leaks, fire outbreaks, and navigation obstacles, and immediately trigger alerts to plant security or emergency services. By using an ESP32-based controller with Wi-Fi capabilities, the data is sent to a PHP-based web application, enabling remote monitoring from any location. The solution is scalable and adaptable for multiple sectors, including chemical plants, manufacturing units, warehouses, and energy facilities. Overall, it contributes to smart industrial systems, aiming for improved safety regulation, accident prevention, and enhanced emergency response.

CHAPTER-2

LITERATURE REVIEW

2.1 LITERATURE COLLECTION

Rehman et al. [1] – “Design and Implementation of IoT based Vehicle Tracking System” (2020), IEEE Proceedings. This research provides a comprehensive framework for real-time fleet management and passenger safety using the ESP8266 and ESP32 microcontrollers. The study focuses on the integration of Neo-6M GPS modules to capture precise longitudinal and latitudinal coordinates, which are then processed to calculate velocity and estimated time of arrival (ETA). By utilizing Firebase as a real-time database, the authors demonstrated a system that can update location data every few seconds with minimal latency. The paper concludes that such IoT-based tracking significantly reduces operational costs for transport companies and provides a layer of security for passengers through "share-location" features and geofencing capabilities.

Gomez-Rosal et al. [2] – “Evaluation of a Smart Mobile Robotic System for Industrial Plant Inspection and Supervision” (2024), IEEE Sensors Journal. This high-impact study evaluates an autonomous mobile robot (AMR) specifically engineered for high-risk industrial environments such as chemical refineries and power plants. The robotic platform utilizes a multi-modal sensor suite including Long-Wave Infrared (LWIR) cameras for thermal anomaly detection and metal-oxide (MOX) gas sensors for chemical leak identification. The research highlights the implementation of a Simultaneous Localization and Mapping (SLAM) algorithm that allows the robot to navigate through complex, unmapped industrial layouts without human intervention. The findings indicate that the robot achieved a 98% success rate in identifying hazardous leaks, proving that automated supervision can drastically minimize human exposure to toxic substances and extreme temperatures.

J. S. Dikgale et al. [3] – “AI-Based Smart System for Accident Detection and Prevention on Curved Roads” (2020), IEEE Conference on Energy Reports. This paper addresses the specific challenges of road safety in geographically difficult terrains, such as mountain passes and sharp curves. The authors propose a sensor-fusion approach combining ultrasonic distance sensors with MPU6050 accelerometers to monitor vehicle stability and proximity to road edges. By employing a lightweight machine learning model on the edge device, the system can predict the likelihood of a rollover or collision before it occurs. The study emphasizes that real-time auditory warnings to the driver, coupled with automated SMS alerts to emergency services via GSM, can reduce fatality rates by ensuring medical assistance reaches the site within the "golden hour" of trauma care.

A. Kumar et al. [4] – “Intelligent IoT Based Automatic Vehicle Speed Control and Alert System” (2025), IEEE Xplore. Focusing on the concept of Smart Cities, this 2025 study introduces an automated speed regulation system that interfaces directly with a vehicle's Electronic Control Unit (ECU). The system uses Radio Frequency Identification (RFID) and GPS-based zone detection to identify "Silent Zones" or "High-Risk Zones" like hospital perimeters and school crossings. Upon entering these zones, the ESP32 controller sends a PWM signal to the motor driver to limit the maximum speed to a pre-defined safety threshold. The research demonstrates that this "electronic governor" approach significantly reduces pedestrian accidents and ensures strict adherence to traffic laws without requiring constant police surveillance.

S. Khan et al. [5] – “Driver Drowsiness Detection Using Swin Transformer and Diffusion Models” (2025), IEEE Transactions on Intelligent Transportation Systems. This state-of-the-art research presents a non-intrusive driver monitoring system (DMS) that utilizes a high-definition camera and a Swin Transformer-based deep learning architecture. The system monitors facial landmarks to calculate the Eye

Aspect Ratio (EAR) and Mouth Aspect Ratio (MAR) to detect signs of micro-sleep and frequent yawning. The novelty of this work lies in its ability to function accurately under varying light conditions—from bright sunlight to total darkness—using infrared illumination. When drowsiness is detected, the system not only alerts the driver but also uploads a short video snippet of the event to a cloud server for fleet manager review, providing a robust tool for commercial driver accountability and accident prevention.

CHAPTER - 3

SOFTWARE AND HARDWARE REQUIREMENTS

3.1 SOFTWARE TOOLS

- 1 Development Device - PC with Windows 7 OS
- 2 C Variant - Embedded 'C'
- 3 IDE - Arduino IDE
- 4 Compiler - AVR- gcc
- 5 Programmer - Inbuilt ICSP Programmer

3.1.1 Development Device – PC with Windows OS The development of the Smart Mobile Robotic System for industrial inspection was conducted on a personal computer running a Windows-based operating system. This device serves as the central workstation for the entire development lifecycle, from initial algorithm design to the final firmware deployment. The PC provides the necessary graphical user interface to manage complex development environments and serves as the bridge for uploading code to the robotic controller. It is responsible for hosting the USB communication drivers, serial monitoring tools, and the integrated development environments required to program the ESP32 microcontroller and manage the PHP-based web application.

3.1.2 C Variant – Embedded 'C' The logic governing the autonomous navigation and sensor processing of the robotic system is written in Embedded C. This variant of the C language is specifically optimized for hardware-level control in resource-constrained environments like microcontrollers. In this project, Embedded C allows for low-level access to the ESP32 registers, facilitating real-time responses to critical

events such as flame detection or gas leak alerts. The language was chosen for its high execution speed, efficiency in memory management, and its ability to handle deterministic timing—an essential factor for the robot's obstacle avoidance and navigation mapping routines.

3.1.3 IDE – Arduino IDE Arduino IDE was the primary software interface used for writing, compiling, and uploading the robotic control firmware. It offers a robust ecosystem of libraries that simplify the integration of specialized industrial sensors such as gas, flame, and ultrasonic modules. The IDE's built-in serial monitor and plotter are critical for debugging analog sensor values in real-time, allowing the developer to calibrate the robot's sensitivity to environmental hazards. Furthermore, the seamless integration with the ESP32 board manager ensures a rapid prototyping cycle and stable firmware performance during field testing.

3.1.4 Compiler – GCC Toolchain For this robotic system, the code is compiled using the Xtensa-esp32-elf-gcc compiler, which is a specialized derivative of the GNU Compiler Collection (GCC). This toolchain is responsible for translating the Embedded C source code into machine language that the ESP32's dual-core processor can execute. The compiler optimizes the code for high performance and minimal memory usage, ensuring that the autonomous navigation algorithms and wireless data transmission protocols run concurrently without system lag or hardware bottlenecks.

3.1.5 Programmer – Inbuilt USB-to-Serial Programmer The ESP32 microcontroller has a built-in USB-to-Serial converter, allowing firmware to be uploaded directly using a standard USB cable without additional programming hardware. It supports automatic reset and easy bootloader access, which simplifies development and testing. This makes updating the robot's control and navigation system faster and more convenient during operation.

3.2 HARDWARE TOOLS

3.2.1 Target Device – ESP32 Microcontroller The ESP32 acts as the "brain" of the smart mobile robotic system. It is a powerful, low-cost System-on-a-Chip (SoC) that includes integrated Wi-Fi and dual-mode Bluetooth for industrial connectivity. Its dual-core processor is leveraged to handle multi-tasking: one core manages the high-speed navigation and sensor data acquisition (gas, flame, and ultrasonic), while the second core handles the transmission of safety alerts to the cloud server. Its ultra-low power consumption modes and wide range of GPIO pins make it the ideal controller for a battery-powered mobile robot operating in hazardous plant zones.

3.2.2 Oscilloscope – Tektronix DSO (Digital Storage Oscilloscope) A Tektronix DSO was utilized during the hardware development phase to visualize electrical signals and verify the integrity of high-frequency communications. It was instrumental in analyzing the UART data lines between the GPS and ESP32, as well as verifying the PWM signals used for the robot's motor control and steering. By capturing and analyzing signal waveforms, the oscilloscope ensured that sensor data was accurate and that the communication between various modules was free from electromagnetic interference common in industrial environments.

3.2.3 Multimeter – Mastech Digital Multimeter The Mastech digital multimeter was an essential tool for verifying the electrical health of the robotic system. It was used to measure voltage levels across the battery and motor drivers, ensuring that the robot's power distribution remained stable under heavy loads. Additionally, the multimeter was used for continuity testing during the circuit assembly phase to prevent short circuits and to validate the calibration of analog gas and flame sensors. This tool ensured that all hardware components operated within their safe and specified electrical parameters before the robot was deployed for inspection tasks.

CHAPTER-4

SYSTEM ANALYSIS

4.1 EXISTING SYSTEM

The current industrial safety paradigm relies heavily on stationary monitoring systems and periodic manual inspections to manage plant environments. These traditional methods lack an efficient mechanism for continuously monitoring the dynamic conditions of large-scale industrial facilities. Without an autonomous mobile tracking system, it becomes extremely difficult to assess the real-time status of distant or high-risk zones, leading to undetected gas leaks, structural stresses, or fire hazards that may escalate into catastrophic failures. Safety officers and plant managers are often unaware of localized environmental shifts until they trigger broad alarms, resulting in delayed corrective maintenance and increased risks to personnel. Additionally, manual supervision is inherently limited by human physical constraints and exposure risks; inspectors cannot remain in hazardous zones for extended periods to monitor fluctuating gas levels or thermal anomalies. Without a robotic system to perform these high-frequency checks, there is no way to provide early predictive warnings or take autonomous preventive measures to ensure the integrity of the facility. Another major limitation is the fragmentation of data reporting. Since there is no continuous mobile data acquisition, authorities are unable to obtain a comprehensive, real-time heat map of the plant's operational health. This lack of centralized, live data makes it difficult to implement proactive safety protocols. Moreover, in the event of a sudden leak or fire, the absence of an automated, mobile alert system causes significant delays in emergency response, as fixed sensors may be positioned too far from the source to provide immediate localized data.

4.1.1 DRAWBACKS

1. Lack of Continuous Industrial Environment Monitoring Industrial facilities are not continuously monitored in every sector, making it difficult to assess the real-time status of remote or hazardous areas. This lack of visibility leads to undetected equipment failures and significantly increases safety risks for the workforce.

2. High Risk in Manual Inspection Tasks Traditional methods rely on human personnel to enter potentially toxic or flammable zones for routine checks. This exposure increases the likelihood of health complications or injuries due to delayed response or the inability to detect odorless gases manually.

3. Absence of Real-Time Mobile Data Reporting Current stationary systems do not support the dynamic tracking of environmental shifts across the entire plant layout. This limits facility managers from taking proactive measures and results in "blind spots" where hazards can develop unnoticed.

4. No Automated Mobile Emergency Response In the event of a localized accident or leak, there is no autonomous system to immediately reach the site and provide live data. This results in delayed emergency intervention, potentially leading to severe infrastructure damage or loss of life.

4.2 PROPOSED SYSTEM

This project introduces an advanced smart mobile robotic system designed to revolutionize industrial plant supervision and safety. The proposed system continuously patrols industrial environments, ensuring a real-time assessment of potential risks through a mobile platform. By integrating state-of-the-art gas sensors, flame sensors, and autonomous navigation modules, the robot provides instant updates on the plant's atmospheric and thermal conditions. The collected data is transmitted via high-speed

wireless protocols to a centralized remote platform for continuous analysis. To ensure comprehensive supervision, the robot includes dedicated modules for obstacle avoidance and environment mapping, allowing it to navigate complex industrial terrains without human intervention. The system identifies signs of gas leaks or fire outbreaks at the source and triggers immediate alerts to prevent large-scale disasters. For anomaly detection, the robot utilizes impact and vibration sensors to identify mechanical irregularities in plant machinery. When a hazard is detected, the system automatically broadcasts the exact location coordinates and sensor readings to emergency contacts and safety dashboards. This proactive solution significantly improves industrial safety by ensuring that critical information is accessible in real-time, allowing for swift intervention and a measurable reduction in plant downtime.

4.2.1 APPLICATIONS

Industrial Plant Supervision Enables facility managers to monitor hazardous production zones and chemical storage areas in real-time. This reduces the need for human patrols in high-risk environments and improves overall operational oversight.

Hazardous Leak Detection Provides specialized monitoring for gas leaks and chemical spills in refineries and processing units. The system identifies leaks at an early stage, preventing potential explosions and ensuring environmental compliance.

Emergency Response Enhancement Acts as a first responder by providing real-time visual and sensor data from the site of an incident. This reduces the time taken for emergency teams to assess the situation and increases the safety of the intervention.

Predictive Facility Maintenance Assists maintenance teams by providing accurate thermal and vibration data of heavy machinery. This allows for the

identification of wear and tear before a breakdown occurs, optimizing the plant's maintenance schedule.

Smart Warehouse Monitoring Integrates with logistics infrastructure to monitor environmental conditions in large storage facilities. It ensures that perishable or sensitive goods are kept within safe temperature and gas concentration limits.

Remote Security and Surveillance Allows authorized personnel to remotely view live feeds and sensor reports from any location. This streamlines facility supervision and ensures that security protocols are maintained even during off-hours or plant shutdowns.

CHAPTER 5

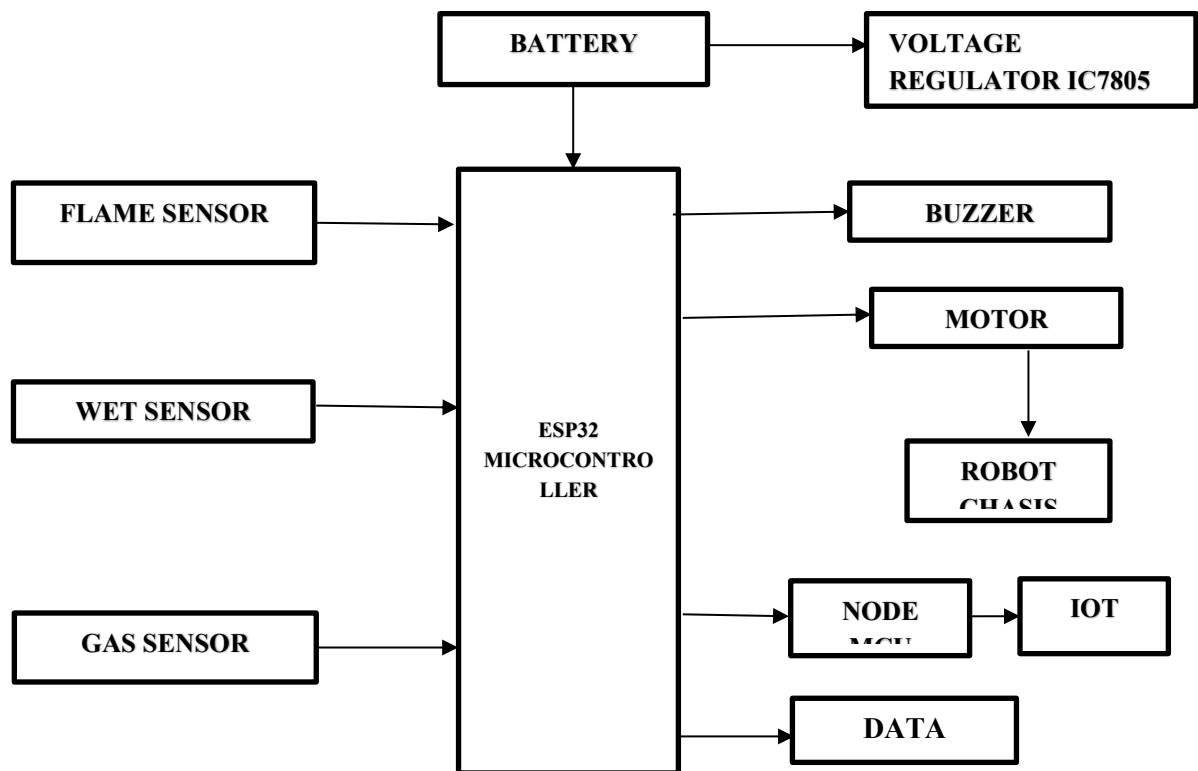
METHODOLOGY

5.1 WORKING FUNCTION

The methodology of this system is centered around the ESP32 microcontroller, which serves as the primary processing hub for the mobile robotic platform and its integrated sensor array. The system operates by continuously polling data from specialized industrial sensors to ensure comprehensive environmental supervision. A flame sensor is utilized to identify the presence of fire by detecting specific infrared wavelengths, while the MQ-series gas sensor monitors the atmosphere for hazardous leaks such as LPG, smoke, or alcohol vapors. For navigation and safety, an ultrasonic sensor is employed to detect obstacles in the robot's path, allowing the ESP32 to calculate alternative routes and avoid collisions in dense plant layouts. A vibration sensor is integrated to monitor the robot's own structural integrity and detect sudden impacts or mechanical anomalies in nearby machinery.

The robot uses a GPS module for real-time location tracking during industrial inspection. The ESP32 controller processes sensor data and responds quickly when abnormal conditions are detected. The system transmits monitoring data wirelessly for real-time observation and can send alert messages during emergencies. A regulated power supply ensures stable and reliable operation of the robot and its components.

5.1.1 BLOCK DIAGRAM



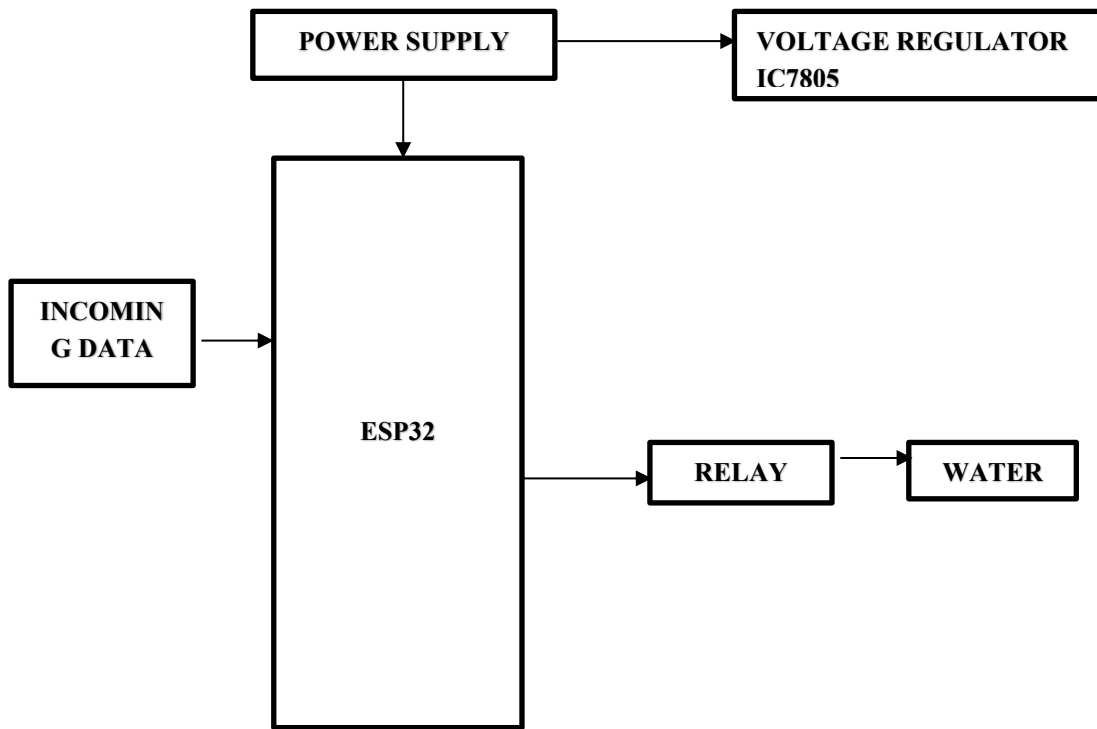


FIG 5.1 BLOCK DIAGRAM

5.2 ARCHITECTURE COMPONENT

5.2.1 POWER SUPPLY UNIT

Power supply is a reference to a source of electrical power. A device or system that supplies electrical or other types of energy to an output load or group of loads is called a power supply unit or PSU. The term is most commonly applied to electrical energy supplies, less often to mechanical ones, and rarely to others.

Power supplies for electronic devices are mainly classified into linear power supplies and switching power supplies. Linear power supplies have a simple design but are bulky, heavy, and less efficient for high-current applications. In contrast, switching power supplies are smaller, lighter, and more efficient, making them suitable for modern electronic systems despite their complex design.

LINEAR POWER SUPPLY

An AC powered linear power supply usually uses a transformer to convert the voltage from the wall outlet (mains) to a different, usually a lower voltage. If it is used to produce DC, a rectifier is used. A capacitor is used to smooth the pulsating current from the rectifier. Some small periodic deviations from smooth direct current will remain, which is known as ripple. These pulsations occur at a frequency related to the AC power frequency (for example, a multiple of 50 or 60 Hz).

The voltage produced by an unregulated power supply will vary depending on the load and on variations in the AC supply voltage. For critical electronics applications a linear regulator will be used to stabilize and adjust the voltage. This regulator will also greatly reduce the ripple and noise in the output direct current. Linear regulators often provide current limiting, protecting the power supply and attached circuit from over current.

Adjustable linear power supplies are common laboratory and service shop test equipment, allowing the output voltage to be set over a wide range. For example, a bench power supply used by circuit designers may be adjustable up to 30 volts and up to 5 amperes output. Some can be driven by an external signal, for example, for applications requiring a pulsed output.

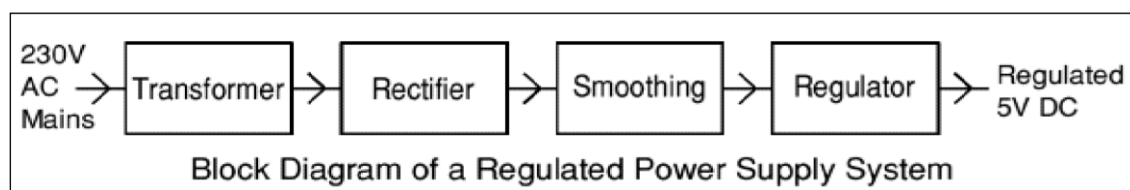


FIG: 5.2.1.1 POWER SUPPLY CIRCUIT

TRANSFORMER

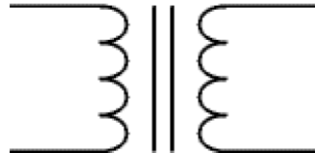


FIG: 5.2.1.2 TRANSFORMER SYMBOL

Transformers convert AC electricity from one voltage to another with little loss of power. Transformers work only with AC and this is one of the reasons why mains electricity is AC.

Step-up transformers increase voltage, step-down transformers reduce voltage. Most power supplies use a step-down transformer to reduce the dangerously high mains voltage (230V in UK) to a safer low voltage.

The input coil is called the primary and the output coil is called the secondary. There is no electrical connection between the two coils; instead they are linked by an alternating magnetic field created in the soft-iron core of the transformer. The two lines in the middle of the circuit symbol represent the core.

Transformers waste very little power so the power out is (almost) equal to the power in. Note that as voltage is stepped down current is stepped up.

The ratio of the number of turns on each coil, called the turn's ratio, determines the ratio of the voltages. A step-down transformer has a large number of turns on its primary (input) coil which is connected to the high voltage mains supply, and a small number of turns on its secondary (output) coil to give a low output voltage.

Turns ratio= $V_p/V_s=N_n/N_s$ and Power out=Power in

$$V_s \cdot I_s = V_p \cdot I_p$$

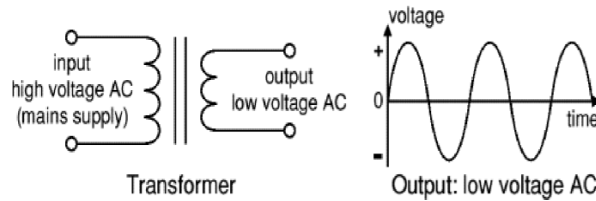


FIG: 5.2.1.3 TRANSFORMER OUTPUT VOLTAGE

RECTIFIER:

There are several ways of connecting diodes to make a rectifier to convert AC to DC. The bridge rectifier is the most important and it produces full-wave varying DC. A full-wave rectifier can also be made from just two diodes if a centre-tap transformer is used, but this method is rarely used now that diodes are cheaper. A single diode can be used as a rectifier but it only uses the positive (+) parts of the AC wave to produce halfwave varying DC.

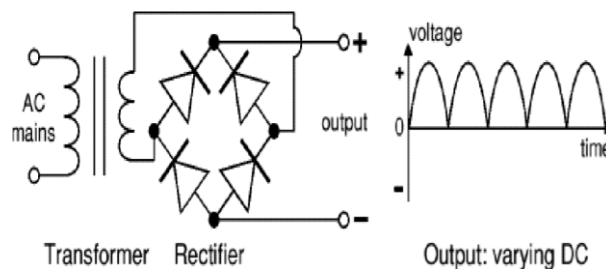


FIG: 5.2.1.4 RECTIFIER CIRCUIT

The varying DC output is suitable for lamps, heaters and standard motors. It is not suitable for electronic circuits unless they include a smoothing capacitor.

BRIDGE RECTIFIER:

A bridge rectifier can be made using four individual diodes, but it is also available in special packages containing the four diodes required. It is called a full-wave rectifier because it uses the entire AC wave (both positive and negative sections). 1.4V is used up in the bridge rectifier because each diode uses 0.7V when conducting and there are always two diodes conducting, as shown in the diagram below.

Bridge rectifiers are rated by the maximum current they can pass and the maximum reverse voltage they can withstand (this must be at least three times the supply RMS voltage so the rectifier can withstand the peak voltages). Please see the Diodes page for more details, including pictures of bridge rectifiers.

Alternate pairs of diodes conduct, changing over the connections so the alternating directions of AC are converted to the one direction of DC.

Output: full-wave varying DC: (using the entire AC wave):

REGULATOR:

Voltage regulator ICs are available with fixed (typically 5, 12 and 15V) or variable output voltages. They are also rated by the maximum current they can pass. Negative voltage regulators are available, mainly for use in dual supplies. Most regulators include some automatic protection from excessive current ('overload protection') and overheating ('thermal protection').

The LM78XX series of three terminal regulators is available with several fixed output voltages making them useful in a wide range of applications. One of these is local on card regulation, eliminating the distribution problems associated with single point regulation. The voltages available allow these regulators to be used in logic

systems, instrumentation, HiFi, and other solid state electronic equipment. Although designed primarily as fixed voltage regulators these devices can be used with external components to obtain adjustable voltages and current.

Many of the fixed voltage regulator ICs has 3 leads and look like power transistors, such as the 7805 +5V 1A regulator shown on the right. They include a hole for attaching a heat sink if necessary.

1. Positive regulator

1. input pin
2. ground pin
3. output pin

It regulates the positive voltage

2. Negative regulator

1. ground pin
2. input pin
3. output pin

It regulate the negative voltage

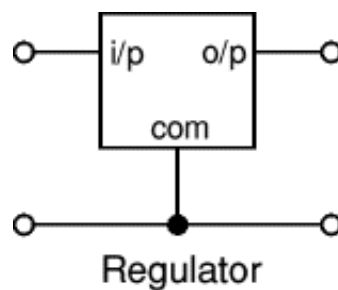


FIG: 5.2.1.5 VOLTAGE REGULATOR

5.2.2 ESP32 MICROCONTROLLER

The ESP32 is a powerful and versatile microcontroller developed by Espressif Systems, widely used in IoT, embedded systems, and smart applications. It is an upgraded version of its predecessor, the ESP8266, and comes with significant enhancements in processing power, connectivity, and peripheral support. The ESP32 is built on a dual-core **or** single-core Tensilica Xtensa LX6 processor, making it suitable for applications that require high computational capability and real-time data processing. It also features built-in Wi-Fi and Bluetooth, enabling seamless wireless communication for smart home systems, industrial automation, and wearable.

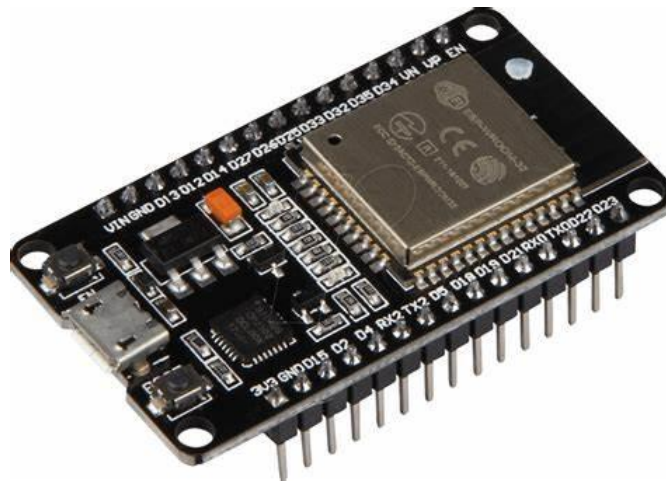


FIG: 5.2.2.1 ESP32 MICROCONTROLLER:

The ESP32 microcontroller boasts impressive hardware specifications that make it a preferred choice for developers and engineers. It operates at a clock speed

of up to 240 MHz and includes up to 520 KB of SRAM and 4 MB of flash memory, ensuring smooth execution of complex tasks. The chip supports Wi-Fi (802.11 b/g/n) for internet connectivity and Bluetooth 4.2 (Classic and Low Energy - BLE) for wireless device communication. Additionally, it comes equipped with a rich set of peripherals, including SPI, I2C, I2S, UART, ADC, DAC, PWM, and touch sensors, allowing it to interface with a wide range of external components and sensors. The ESP32 also integrates cryptographic hardware acceleration, making it ideal for secure communication applications.unstable.

If using more than 12V, the voltage regulator may overheat and damage the board. The recommended range is 7 to 12 volts.

THE POWER PINS ARE AS FOLLOWS:

- **VIN.** The input voltage to the Arduino board when it's using an external power source (as opposed to 5 volts from the USB connection or other regulated power source). You can supply voltage through this pin, or, if supplying voltage via the power jack, access it through this pin.
- **5V.** The regulated power supply used to power the microcontroller and other components on the board. This can come either from VIN via an onboard regulator, or be supplied by USB or another regulated 5V supply.
- **3V3.** A 3.3 volt supply generated by the on-board regulator. Maximum current draw is 50 mA.
- **GND.** Ground pins.

INPUT & OUTPUT:

ESP32 has 34 GPIO pins, which can be configured as input or output using `pinMode()`, `digitalWrite()`, and `digitalRead()` functions.

Unlike Arduino, ESP32 operates at 3.3V logic level. Each GPIO pin supports a maximum current draw of 40 mA. Some GPIOs have specialized functions:

Serial Communication:

- GPIO1 (TX) and GPIO3 (RX): Used for UART serial communication.
- Additional UARTs available on other GPIOs (UART0, UART1, UART2).

External Interrupts:

- All GPIO pins can be configured as interrupts using the attach Interrupt() function, supporting rising, falling, or change in value triggers.
- PWM (Pulse Width Modulation):
- Any GPIO pin can be used for PWM output using the led Write() function. ESP32 supports 16 PWM channels with 8, 10, 12, or 16-bit resolution.

SPI (Serial Peripheral Interface):

VSPI (Default SPI):

- GPIO23 (MOSI), GPIO19 (MISO)
- GPIO18 (SCK), GPIO5 (SS)

HSPI (Alternative SPI):

- Can be assigned to different GPIOs.

I2C (Inter-Integrated Circuit):

- GPIO21 (SDA) and GPIO22 (SCL) support I2C communication using the Wire library.

Built-in LED:

- GPIO2 is typically connected to an on board LED, which can be controlled by setting the pin HIGH (LED on) or LOW (LED off).

Analog Inputs:

- ESP32 has 18 ADC (Analog-to-Digital Converter) channels with 12-bit resolution (4096 different values). By default, they measure voltage from 0V to 3.3V, but this range can be adjusted using the `analogSetAttenuation()` function.

Some key ADC pins include:

- GPIO32–GPIO39: ADC1 (Preferred for Wi-Fi applications).
- GPIO0, GPIO2, GPIO4, GPIO12–GPIO15, GPIO25–GPIO27: ADC2 (Avoid using when Wi-Fi is active).

Other Pins:**DAC (Digital-to-Analog Converter):**

- GPIO25 & GPIO26 provide 8-bit DAC output, allowing direct analog signal generation.

AREF (Analog Reference):

- Not present in ESP32, but attenuation settings allow range adjustments.

Reset:

- EN (Enable Pin) resets the microcontroller when pulled LOW.

Touch Sensors:

- ESP32 includes 10 capacitive touch GPIOs (e.g., GPIO4, GPIO12, GPIO13, GPIO14, etc.) for touch-based input applications.

DIMENSION DRAWING:

- The **ESP32 dimension drawing** typically includes measurements for the module's length, width, and pin placements.
- The standard ESP32 development board measures around **51mm × 25mm**, while the ESP32-WROOM-32 module itself is approximately **18mm × 25.5mm**.
- The drawing details the **GPIO pin arrangement**, **antenna positioning**, and **mounting hole locations** to assist with PCB design and hardware integration.

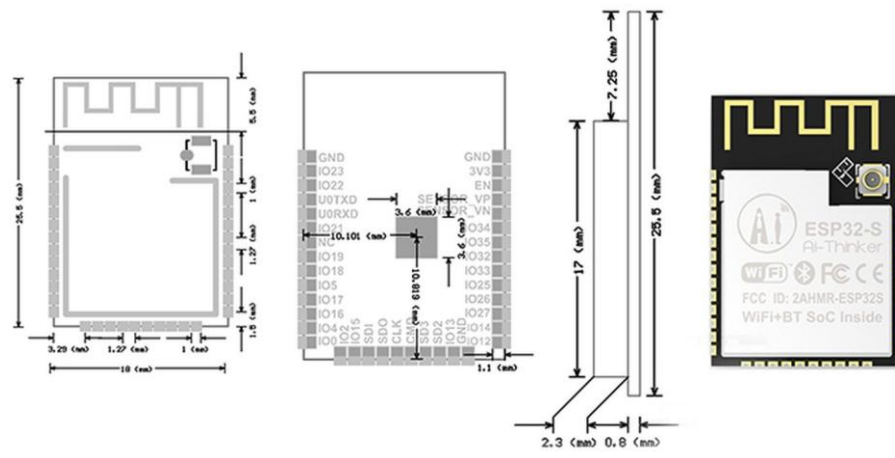


FIG: 5.2.2.2 DIMENSION DRAWING

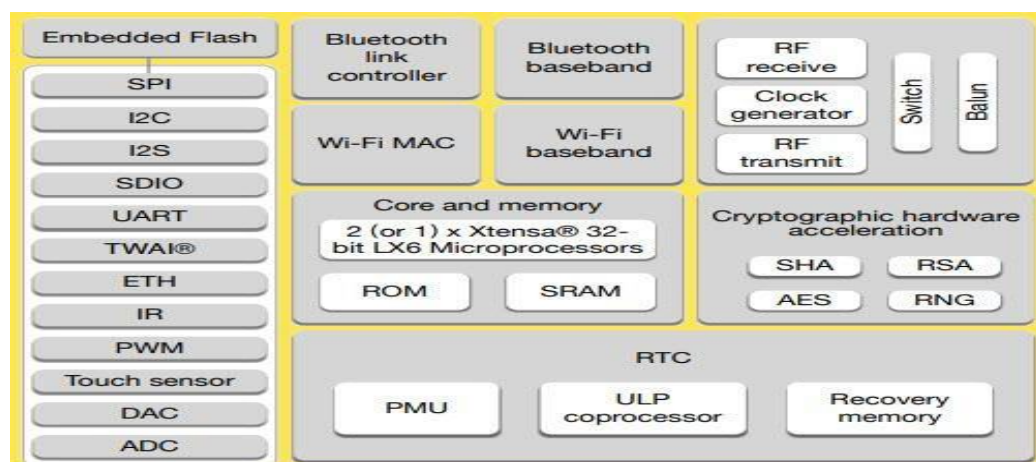


FIG: 6.2.2.3 MEMORY BITS

SPECIFICATIONS OF ESP32

ESP32 has a lot more features than ESP8266 and it is difficult to include all the specifications in this Getting Started with ESP32 guide. So, I made a list of some of the important specifications of ESP32 here. But for complete set of specifications, I strongly suggest you to refer to the Datasheet.

- Single or Dual-Core 32-bit LX6 Microprocessor with clock frequency up to 240 MHz.
- 520 KB of SRAM, 448 KB of ROM and 16 KB of RTC SRAM.
- Supports 802.11 b/g/n Wi-Fi connectivity with speeds up to 150 Mbps.
- Support for both Classic Bluetooth v4.2 and BLE specifications.
- 34 Programmable GPIOs.
- Up to 18 channels of 12-bit SAR ADC and 2 channels of 8-bit DAC □ Serial Connectivity include 4 x SPI, 2 x I²C, 2 x I²S, 3 x UART.
- Ethernet MAC for physical LAN Communication (requires external PHY).
- 1 Host controller for SD/SDIO/MMC and 1 Slave controller for SDIO/SPI.
- Motor PWM and up to 16-channels of LED PWM.
- Secure Boot and Flash Encryption.
- Cryptographic Hardware Acceleration for AES, Hash (SHA-2), RSA, ECC and RNG.

DIFFERENT WAYS TO PROGRAM

A good hardware like ESP32 will be more user friendly if it can be programmed (writing code) in more than one way. And not surprisingly, the ESP32 supports multiple programming environments.

Some of the commonly used programming environments are:

- Arduino IDE
- PlatformIO IDE (VS Code)
- LUA
- MicroPython
- Espressif IDF (IoT Development Framework)
- JavaScript

As Arduino IDE is already a familiar environment, we will use the same to program ESP32 in our upcoming projects. But you can definitely try out others as well.

LAYOUT

We will see what a typical ESP32 Development Board consists of by taking a look at the layout of one of the popular low-cost ESP Boards available in the market called the ESP32 DevKit Board.

IMPORTANT NOTE:

There are many ESP32 Boards based on ESP-WROOM-32 Module available in the market. The layout, pinout and features vary from board to board. The board which I have has 30 Pins (15 pins on each side). There are some board with 36 Pins and some with slightly less Pins. So, double check the pins before making connections or even powering up the board.

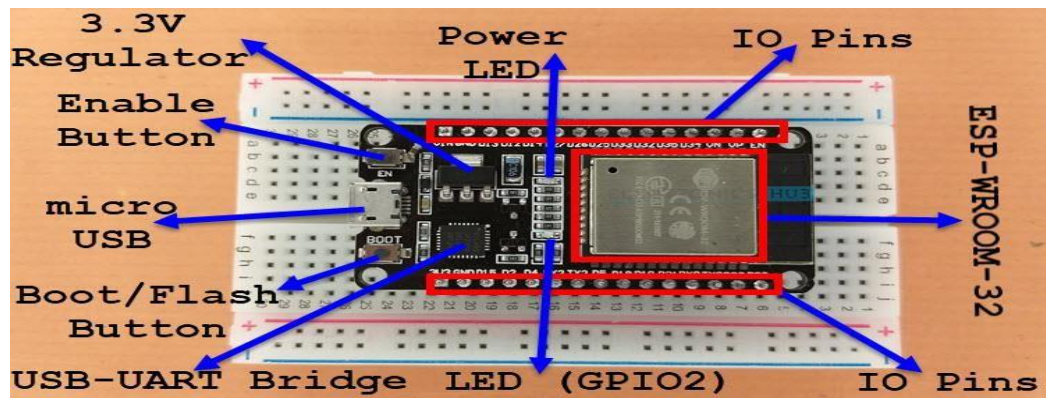


FIG: 5.2.2.4 POWER PINS

As you can see from the image, the ESP32 Board consists of the following:

- ESP-WROOM-32 Module
- Two rows of IO Pins (with 15 pins on each side)
- CP2012 USB – UART Bridge IC
- micro–USB Connector (for power and programming)
- AMS1117 3.3V Regulator IC
- Enable Button (for Reset)

- Boot Button (for flashing)
- Power LED (Red)
- User LED (Blue – connected to GPIO2)
- Some passive components

An interesting point about the USB-to-UART IC is that its DTR and RTS pins are used to automatically set the ESP32 in to programming mode (whenever required) and also reset the board after programming.

PINOUT OF ESP32 BOARD

- I will make a separate dedicated tutorial on ESP32 Pinout. But for the time being, take a look the pinout diagram of the ESP32 Development Board.
- The ESP32 offers multiple power modes, including deep sleep and light sleep, allowing efficient power consumption for battery-operated applications.

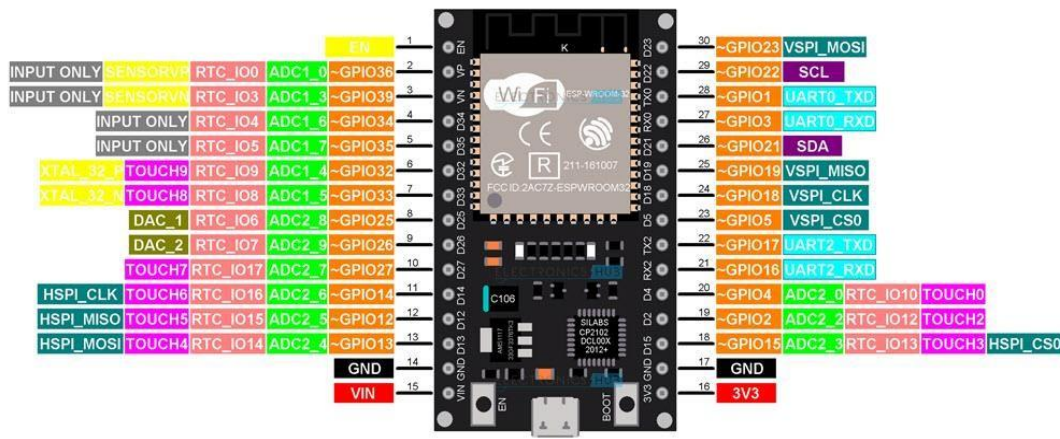


FIG: 5.2.2.5 ESP32 PINS

5.2.3 WET SENSOR

FEATURES:

- Operating voltage: 5V
- Provide both digital and analog output
- Adjustable sensitivity
- Output LED indicator
- Compatible with Arduino
- TTL Compatible
- Bolt holes for easy installation
- Jumper wires included

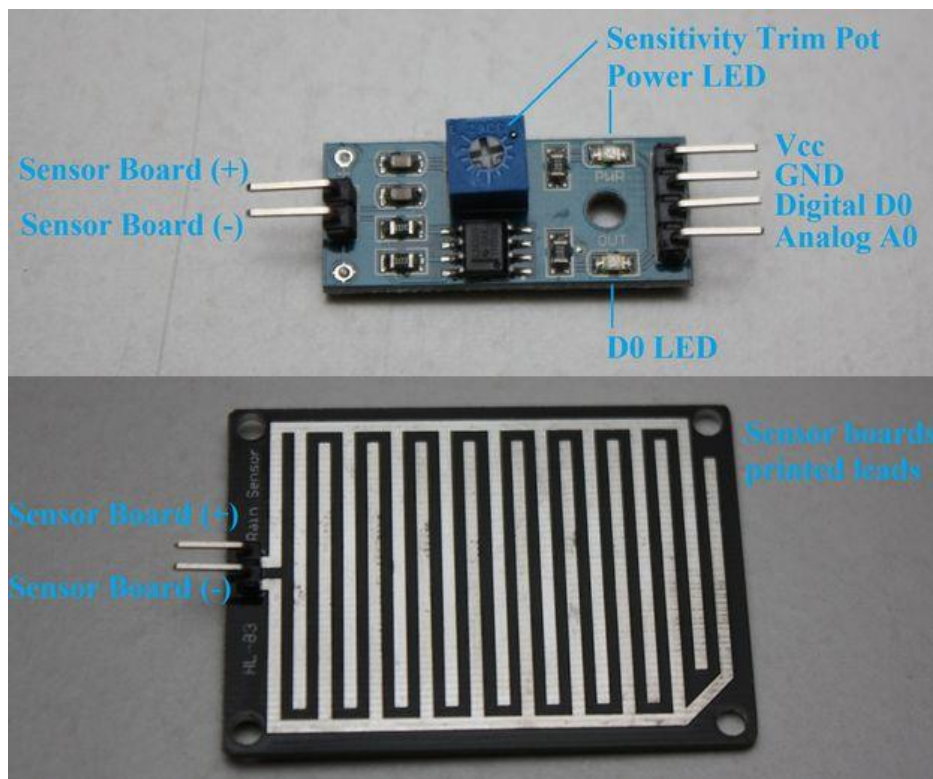


Fig: 5.2.3.1 Wet Sensor

USAGE:

Rain sensors are used in the detection of water beyond what a humidity sensor can detect.

HOW IT WORKS:

The rain sensor detects water that completes the circuits on its sensor boards' printed leads. The sensor board acts as a variable resistor that will change from 100k ohms when wet to 2M ohms when dry. In short, the wetter the board the more current that will be conducted.

PINS:

A0..... Analog output

D0..... Digital output

GND..... Ground

VCC..... Positive voltage (input: 5v for analog 3.3v for Digital.)

LOOP PINS:

+ Sensor board hookup A

- Sensor board hookup B

Material	Plastic + PCB + iron
Specification	Voltage: DC 5V; Power indicator, output LED indicator; TTL output; TTL output effective signal is low level; Drive capability: about 100mA; Can directly drive relay, buzzer, fan; The sensitivity can be adjusted by potentiometer; The LED will light up the high level as no rain; As rains, output low level, LED is on

Features	Sensor
Application	DIY project and other electronic products
Dimensions	2.17 in x 1.57 in x 0.08 in (5.5 cm x 4.0 cm x 0.2 cm) Weight: 0.25 oz (7 g)

5.2.4 GAS SENSOR

A **gas detector** is a device that detects the presence of gases in an area, often as part of a safety system. A gas detector can sound an alarm to operators in the area where the leak is occurring, giving them the opportunity to leave. This type of device is important because there are many gases that can be harmful to organic life, such as humans or animals.

Gas detectors can be used to detect combustible, flammable and toxic gases, and oxygen depletion. This type of device is used widely in industry and can be found in locations, such as on oil rigs, to monitor manufacturing processes and emerging technologies such as photovoltaic. They may be used in firefighting.

Gas leak detection is the process of identifying potentially hazardous gas leaks by sensors. Additionally a visual identification can be done using a thermal camera. These sensors usually employ an audible alarm to alert people when a dangerous gas has been detected. Exposure to toxic gases can also occur in operations such as painting, fumigation, fuel filling, construction, excavation of contaminated soils, landfill operations, entering confined spaces, etc. Common sensors include combustible gas sensors, photoionization detectors, infrared point sensors, ultrasonic sensors, electrochemical gas sensors, and metal–oxide–semiconductor (MOS) sensors. More recently, infrared imaging sensors have come into use. All of these sensors are used for a wide range of applications and can be found in industrial plants, refineries, pharmaceutical manufacturing, fumigation facilities, paper pulp mills, aircraft and

shipbuilding facilities, hazmat operations, waste-water treatment facilities, vehicles, indoor air quality testing and homes.

Gas leak detection methods became a concern after the effects of harmful gases on human health were discovered.

Before modern electronic sensors, early detection methods relied on less precise detectors. Through the 19th and early 20th centuries, coal miners would bring canaries down to the tunnels with them as an early detection system against life-threatening gases such as carbon dioxide, carbon monoxide and methane. The canary, normally a very songful bird, would stop singing and eventually die if not removed from these gases, signaling the miners to exit the mine quickly.

The first gas detector in the industrial age was the *flame safety lamp* (or Davy lamp) was invented by Sir Humphry Davy (of England) in 1815 to detect the presence of methane (firedamp) in underground coal mines. The flame safety lamp consisted of an oil flame adjusted to specific height in fresh air. To prevent ignition with these lamps the flame was contained within a glass sleeve with a mesh flame arrestor. The flames height varied depending on the presence of methane (higher) or the lack of oxygen (lower). To this day, in certain parts of the world flame safety lamps are still in service.

The modern era of gas detection started in 1926–1927 with the development of the catalytic combustion (LEL) sensor by Dr. Oliver Johnson. Dr Johnson was an employee of Standard Oil Company in California (now Chevron), he began research and development on a method to detect combustible mixtures in air to help prevent explosions in fuel storage tanks. A demonstration model was developed in 1926 and denoted as the Model A. The first practical "electric vapor indicator" meter begun production in 1927 with the release of the Model B.

TYPES:

Gas detectors can be classified according to the operation mechanism (semiconductor, oxidation, catalytic, photoionization, infrared, etc.). Gas detectors come packaged into two main form factors: portable devices and fixed gas detectors.

Portable detectors are used to monitor the atmosphere around personnel and are either hand-held or worn on clothing or on a belt/harness. These gas detectors are usually battery operated. They transmit warnings via audible and visible signals, such as alarms and flashing lights, when dangerous levels of gas vapors are detected.

Fixed type gas detectors may be used for detection of one or more gas types. Fixed type detectors are generally mounted near the process area of a plant or control room, or an area to be protected, such as a residential bedroom. Generally, industrial sensors are installed on fixed type mild steel structures and a cable connects the detectors to a supervisory control and data acquisition (SCADA) system for continuous monitoring. A tripping interlock can be activated for an emergency situation.



Fig : 5.2.4.1 Gas Sensor

ELECTROCHEMICAL GAS

Electrochemical gas detectors work by allowing gases to diffuse through a porous membrane to an electrode where it is either chemically oxidized or reduced. The amount of current produced is determined by how much of the gas is oxidized at the electrode,^[4] indicating the concentration of the gas. Manufacturers can customize electrochemical gas detectors by changing the porous barrier to allow for the detection of a certain gas concentration range. Also, since the diffusion barrier is a physical/mechanical barrier, the detectors tend to be more stable and reliable over the sensor's duration and thus required less maintenance than other early detector technologies.

However, the sensors are subject to corrosive elements or chemical contamination and may last only 1–2 years before a replacement is required.^[5] Electrochemical gas detectors are used in a wide variety of environments such as refineries, gas turbines, chemical plants, underground gas storage facilities, and more.

Infrared point

Infrared (IR) point sensors use radiation passing through a known volume of gas; energy from the sensor beam is absorbed at certain wavelengths, depending on the properties of the specific gas. For example, carbon monoxide absorbs wavelengths of about 4.2-4.5 μm .^[6] The energy in this wavelength is compared to a wavelength outside of the absorption range; the difference in energy between these two wavelengths is proportional to the concentration of gas present.^[6]

This type of sensor is advantageous because it does not have to be placed into the gas to detect it and can be used for remote sensing. Infrared point sensors can be used to detect hydrocarbons^[7] and other infrared active gases such as water vapor and carbon dioxide. IR sensors are commonly found in waste-water

treatment facilities, refineries, gas turbines, chemical plants, and other facilities where flammable gases are present and the possibility of an explosion exists. The remote sensing capability allows large volumes of space to be monitored.

Engine emissions are another area where IR sensors are being researched. The sensor would detect high levels of carbon monoxide or other abnormal gases in vehicle exhaust and even be integrated with vehicle electronic systems to notify drivers.

5.2.5 FIRE SENSOR

A **flame detector** is a sensor designed to detect and respond to the presence of a flame or fire. Responses to a detected flame depend on the installation, but can include sounding an alarm, deactivating a fuel line (such as a propane or a natural gas line), and activating a fire suppression system. When used in applications such as industrial furnaces, their role is to provide confirmation that the furnace is properly lit; in these cases they take no direct action beyond notifying the operator or control system. A flame detector can often respond faster and more accurately than a smoke or heat detector due to the mechanisms it uses to detect the flame.

OPTICAL FLAME DETECTORS

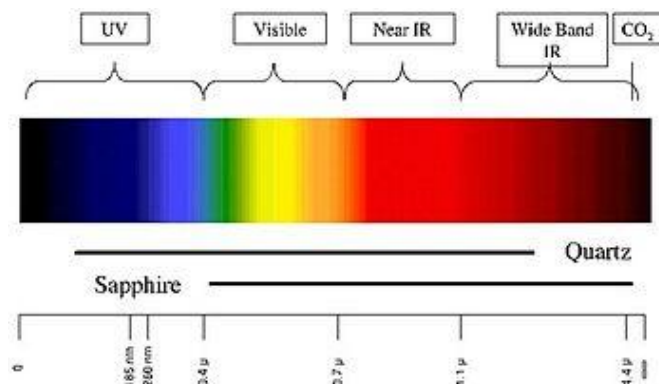


Fig: 5.2.5.1 Flame detector type regions

ULTRAVIOLET

Ultraviolet (UV) detectors work by detecting the UV radiation emitted at the instant of ignition. While capable of detecting fires and explosions within 3–4 milliseconds, a time delay of 2–3 seconds is often included to minimize false alarms which can be triggered by other UV sources such as lightning, arc welding, radiation, and sunlight. UV detectors typically operate with wavelengths shorter than 300 nm. The solar blind UV wavelength band is also easily blinded by oily contaminants.

NEAR IR ARRAY

Near infrared (IR) array flame detectors, also known as visual flame detectors, employ flame recognition technology to confirm fire by analyzing near IR radiation using a charge-coupled device (CCD).

INFRARED

Infrared (IR) flame detectors monitor the infrared spectral band for specific patterns given off by hot gases. These are sensed using a specialized fire-fighting thermal imaging camera (TIC), a type of thermographic camera. False alarms can be caused by other hot surfaces and background thermal radiation in the area. Water on the detector's lens will greatly reduce the accuracy of the detector, as will exposure to direct sunlight. A single-frequency IR flame detector is typically sensitive to wavelengths around 4.4 micrometers, which is a spectral characteristic peak of hot carbon dioxide as is produced in a fire. The usual response time of an IR detector is 3–5 seconds.

UV/IR

These detectors are sensitive to both UV and IR wavelengths, and detect flame by comparing the threshold signal of both ranges. This helps minimize false alarms.

IR/IR FLAME DETECTION

Dual IR (IR/IR) flame detectors compare the threshold signal in two infrared ranges. Often one sensor looks at the 4.4 micrometer carbon dioxide (CO₂) emission, while the other sensor looks at a reference frequency. Sensing the CO₂ emission is appropriate for hydrocarbon fuels; for non-carbon based fuels, e.g., hydrogen, the broadband water bands are sensed.

IR3 FLAME DETECTION

Triple-IR flame detectors compare three specific wavelength bands within the IR spectral region and their ratio to each other. In this case one sensor looks at the 4.4 micrometer range while the other sensors look at reference wavelengths both above and below 4.4. This allows the detector to distinguish between non-flame IR sources and actual flames which emit hot CO₂ in the combustion process. As a result, both detection range and immunity to false alarms can be significantly increased. IR3 detectors can detect a 0.1m²(1 ft²) gasoline pan fire at up to 65 m (215 ft) in less than 5 seconds. Triple IRs, like other IR detector types, are susceptible to blinding by a layer of water on the detector's window.

Most IR detectors are designed to ignore constant background IR radiation, which is present in all environments. Instead they are designed to detect suddenly changing or increasing sources of the radiation. When exposed to changing patterns of non-flame IR radiation, IR and UV/IR detectors become more prone to false alarms, while IR3 detectors become somewhat less sensitive but are more immune to false alarms.

VISIBLE SENSORS

In some detectors, a sensor for visible radiation (light) is added to the design in order to better discriminate against false alarms or to improve the detection range

VIDEO

Closed-circuit television or a web camera can be used for visual detection of (wavelengths between 0.4 and 0.7 μm). Smoke or fog can limit the effective range of these, since they operate solely in the visible spectrum.

OTHER TYPES

IONIZATION CURRENT FLAME DETECTION

The intense ionization within the body of a flame can be measured by means of a current that flows when a voltage is applied by the phenomena of Flame Rectification. This current can be used to verify flame presence and quality. Such detectors are used in large industrial process gas heaters and are connected to the flame control system. They usually act as both flame quality monitors and for flame failure detection. These types of sensors are also common in a variety of household gas furnaces.

THERMOCOUPLE FLAME DETECTION

Thermocouples are used extensively for monitoring flame presence in combustion heating systems and gas cookers. A common use in these installations is to cut off the supply of fuel if the flame fails, in order to prevent unburned fuel from accumulating. These sensors measure heat and therefore are commonly used to determine the *absence* of a flame. This can be used to verify the presence of a Pilot flame.

APPLICATIONS

- UV/IR flame detectors are used in:
- Hydrogen stations.
- Gas-fueled cookers
- Industrial heating and drying systems
- Domestic heating systems
- Industrial gas turbines

5.2.6 BATTERY

In electricity, a battery is a device consisting of one or more electrochemical cells that convert stored chemical energy into electrical energy. Since the invention of the first battery (or "voltaic pile") in 1800 by Alessandro Volta and especially since the technically improved Daniell cell in 1836, batteries have become a common power source for many household and industrial applications. According to a 2005 estimate, the worldwide battery industry generates US\$48 billion in sales each year, with 6% annual growth.

There are two types of batteries: primary batteries (disposable batteries), which are designed to be used once and discarded, and secondary batteries (rechargeable batteries), which are designed to be recharged and used multiple times. Batteries come in many sizes from miniature cells used to power hearing aids and wristwatches to battery banks the size of rooms that provide standby power for telephone exchanges and computer data centers.



Fig: 5.2.6.1 Battery

A battery is a device that converts chemical energy directly to electrical energy. It consists of a number of voltaic cells; each voltaic cell consists of two half-cells connected in series by a conductive electrolyte containing anions and cations. One half-cell includes electrolyte and the electrode to which anions (negatively charged ions) migrate, i.e., the anode or negative electrode; the other half-cell includes electrolyte and the electrode to which cations (positively charged ions) migrate, i.e., the cathode or positive electrode. In the redox reaction that powers the battery, cations are reduced (electrons are added) at the cathode, while anions are oxidized (electrons are removed) at the anode. The electrodes do not touch each other but are electrically connected by the electrolyte. Some cells use two half-cells with different electrolytes. A separator between half-cells allows ions to flow, but prevents mixing of the electrolytes.

Batteries are classified into two broad categories, each type with advantages and disadvantages.

- **Primary batteries** irreversibly (within limits of practicality) transform chemical energy to electrical energy. When the initial supply of reactants is exhausted, energy cannot be readily restored to the battery by electrical means.
- **Secondary batteries** can be recharged; that is, they can have their chemical reactions reversed by supplying electrical energy to the cell, restoring their original composition.

Some types of primary batteries used, for example, for telegraph circuits, were restored to operation by replacing the components of the battery consumed by the chemical reaction. Secondary batteries are not indefinitely rechargeable due to dissipation of the active materials, loss of electrolyte and internal corrosion.

PRIMARY BATTERIES

Primary batteries can produce current immediately on assembly. Disposable batteries are intended to be used once and discarded. These are most commonly used in portable devices that have low current drain, are used only intermittently, or are used well away from an alternative power source, such as in alarm and communication circuits where other electric power is only intermittently available. Disposable primary cells cannot be reliably recharged, since the chemical reactions are not easily reversible and active materials may not return to their original forms. Battery manufacturers recommend against attempting recharging primary cells. Common types of disposable batteries include zinc–carbon batteries and alkaline batteries. In general, these have higher energy densities than rechargeable batteries, but disposable batteries do not fare well under high-drain applications with loads under 75 ohms (75 Ω).

SECONDARY BATTERIES

Secondary batteries must be charged before use; they are usually assembled with active materials in the discharged state. Rechargeable batteries or *secondary cells* can be recharged by applying electric current, which reverses the chemical reactions that occur during its use. Devices to supply the appropriate current are called chargers or rechargers.

A lead–acid battery is one of the oldest types of rechargeable batteries and is widely used due to its low cost and high current supply capacity. It contains liquid electrolyte,

so it must be kept upright and used in a well-ventilated area for safe operation. Although it is heavy compared to other batteries, it remains a common choice for applications that require high power capacity where weight is not a major concern.

5.2.7 WATER MOTOR

Water motors (pumps) are widely used in households, agriculture, and industries for lifting and supplying water from underground sources or tanks. Manual operation often leads to water overflow or dry running, which can damage the motor and waste resources. An automated system ensures efficient water usage and protection of the motor.

Objective

To design and implement a smart control system for a water motor that automatically switches ON/OFF based on water levels, ensuring efficient water management and motor protection.

Working Principle:

The system continuously monitors the water level in the overhead tank using sensors. When the water drops below a minimum level, the microcontroller triggers the relay to turn ON the water motor. Once the tank reaches the maximum level, the motor is automatically turned OFF, preventing overflow.

Specifications:

Component	Specification
Water Motor (Pump)	12V DC Submersible / 0.5 HP AC Motor
Operating Voltage	12V DC / 230V AC (based on motor type)
Water Flow Rate	10–15 L/min (for DC) or 0.5 HP = ~2000 L/h (for AC)

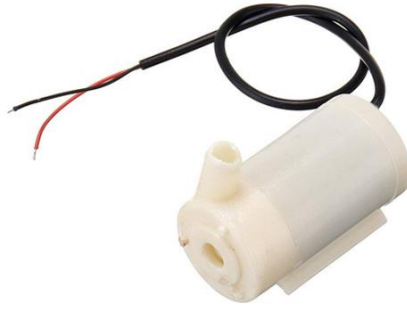


Fig: 5.2.7.1 Water Motor

5.2.8 BUZZER

A **buzzer** or **beeper** is a signalling device, The word "buzzer" comes from the rasping noise that buzzers made when they were electromechanical devices, operated from stepped-down AC line voltage at 50 or 60 cycles. Other sounds commonly used to indicate that a button has been pressed are a ring or a beep

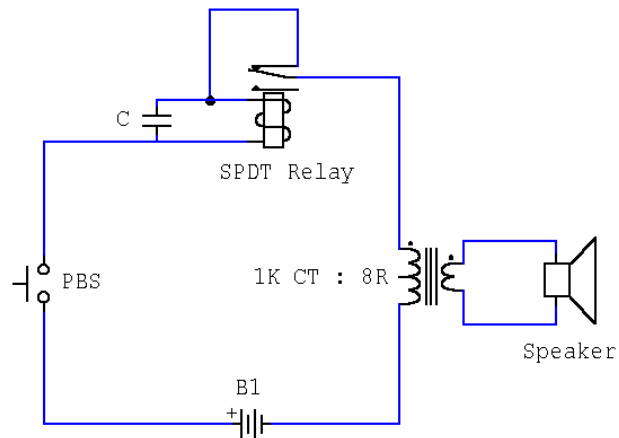


Fig: 5.2.8.1 Buzzer Circuit

This novel buzzer circuit uses a relay in series with a small audio transformer and speaker. When the switch is pressed, the relay will operate via the transformer primary and closed relay contact. As soon as the relay operates the normally closed contact will open, removing power from the relay, the contacts close and the sequence repeats, all very quickly...so fast that the pulse of current causes fluctuations in the transformer primary, and hence secondary.

The speakers tone is thus proportional to relay operating frequency. The capacitor C can be used to "tune" the note. The nominal value is 0.001 μ F, increasing capacitance lowers the buzzers tone.

5.2.9 L293N MOTOR DRIVER

The L293N motor driver is a widely used integrated circuit (IC) designed to control the direction and speed of Gear motors and stepper motors. It is a dual H-Bridge motor driver, meaning it can independently drive two motors in both forward and reverse directions. This makes it an essential component in various motorized applications, such as robotics, automation, and motorized vehicles. Its ability to handle high currents and operate across a broad voltage range makes it compatible with different microcontrollers and embedded systems.

One of the key advantages of the L293N is its built-in safety features, including thermal shutdown protection and flyback diodes. The thermal shutdown ensures that the IC does not overheat during prolonged operation, while the diodes protect against voltage spikes generated by inductive loads such as motors. Additionally, the IC supports Pulse Width Modulation (PWM) control, allowing precise speed and direction adjustments, which is crucial for applications requiring fine motor control.

Due to its ease of use, cost-effectiveness, and reliability, the L293N motor driver is a popular choice among engineers, hobbyists, and developers. It is commonly used in projects involving robotic arms, smart home automation, and industrial machinery. Whether in simple motorized systems or complex autonomous robots, the L293N provides an efficient and stable solution for motor control, making it an integral part of modern electromechanical applications.

Key Features of L293N:

- **Dual H-Bridge Design** – Controls two Gear motors or one stepper motor independently.
- **High Current Output** – Can supply up to **600mA per channel (1.2A peak)** for driving small to medium-sized motors.
- **Wide Voltage Range** – Operates from **4.5V to 36V**, making it suitable for multiple applications.
- **PWM Speed Control** – Allows **variable speed control** using **PWM signals** from microcontrollers.
- **Built-in Flyback Diodes** – Protects the circuit from voltage spikes caused by inductive loads.
- **Thermal Shutdown Protection** – Prevents overheating and damage due to excessive current flow.
- **Enable/Disable Control Pins** – Provides **motor braking and standby modes** for power efficiency.
- **Compact & Cost-Effective** – Easily integrates into electronic circuits due to its **small DIP package**.

Applications of L293N Motor Driver:

- Robotics – Used in autonomous robots and robotic arms for precise movement.
- Motorized Vehicles – Ideal for controlling wheeled robots, RC cars, and drones.
- CNC Machines – Helps in driving stepper motors for precision control.
- Smart Home Automation – Used in motorized doors, curtains, and lifting systems.
- Industrial Automation – Applied in conveyor belts, actuators, and automated machinery.

The L293N motor driver IC is a versatile, reliable, and essential component for motor control applications, making it an excellent choice for engineers, hobbyists, and developers working with motorized systems.

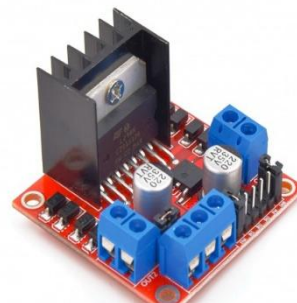
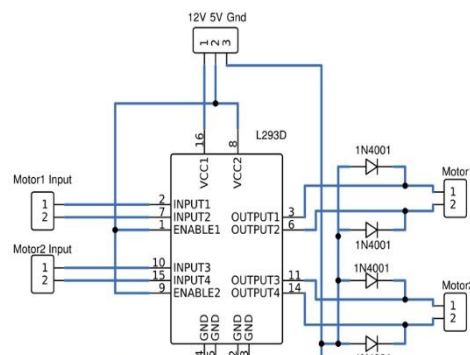


FIG: 5.2.9.1 L293N Circuit

The L293N motor driver circuit is designed to control the direction and speed of Gear motors and stepper motors efficiently. It operates as a dual H-Bridge, allowing independent bidirectional control of two motors using low-power control signals from a microcontroller or other logic circuits.

By utilizing Pulse Width Modulation (PWM) signals, the speed of the motors can be precisely adjusted, making it ideal for robotics, automation, and motorized projects.

The IC can handle high current loads and operates within a wide voltage range, ensuring compatibility with various motor types.

One of the key features of the L293N motor driver circuit is its built-in flyback diodes, which protect the circuit from voltage spikes caused by inductive loads like motors.

Additionally, it includes thermal shutdown and overcurrent protection, ensuring safe operation even under high-stress conditions.

The ease of interfacing with microcontrollers like Arduino and Raspberry Pi makes it a popular choice for engineers and hobbyists.

With its compact design and robust performance, the L293N motor driver circuit plays a crucial role in motorized applications, from simple DIY projects to complex industrial automation systems.

5.2.10 GEAR MOTOR

A Gear motor in simple words is a device that converts direct current (electrical energy) into mechanical energy. It's of vital importance for the industry today, and is equally important for engineers to look into the working principle of Gear motor in details that has been discussed in this article. In order to understand the operating principle of Gear motor we need to first look into its constructional feature.

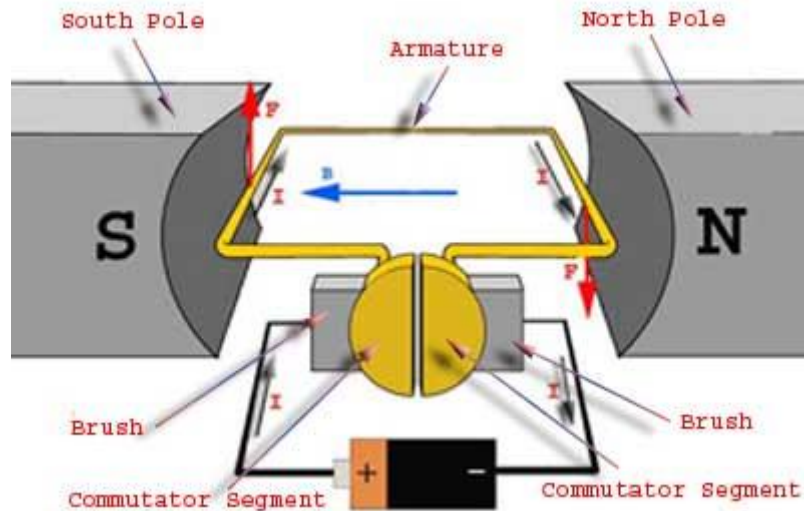


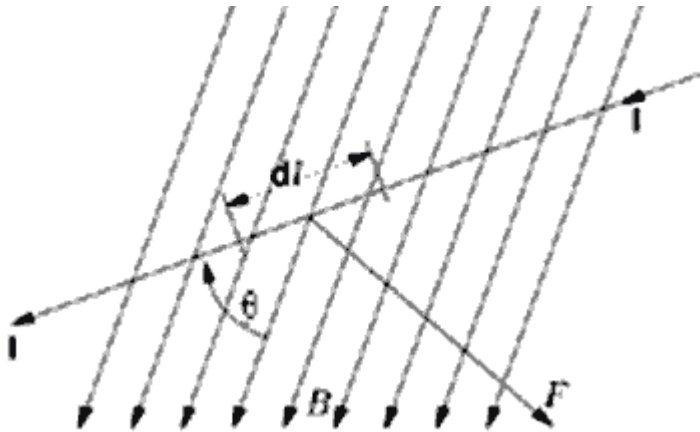
Fig: 5.2.10.1 Gear Motor

The very basic construction of a Gear motor contains a current carrying armature which is connected to the supply end through commutator segments and brushes and placed within the north south poles of a permanent or an electro-magnet as shown in the diagram below.

Now to go into the details of the **operating principle of Gear motor** its important that we have a clear understanding of Fleming's left hand rule to determine the direction of force acting on the armature conductors of Gear motor. Fleming's left hand rule says that if we extend the index finger, middle finger and thumb of our left hand in such a way that the current carrying conductor is placed in a magnetic field (represented by the index finger) is perpendicular to the direction of current (represented by the middle finger), then the conductor experiences a force in the direction (represented by the thumb) mutually perpendicular to both the direction of field and the current in the conductor.

For clear understanding the **principle of Gear motor** we have to determine the magnitude of the force, by considering the diagram below.

We know that when an infinitely small charge dq is made to flow at a velocity 'v' under the influence of an electric field E , and a magnetic field B , then the Lorentz Force dF experienced by the charge is given by:-



$$dF = dq(E + vB)$$

For the **operation of Gear motor**, considering $E = 0$

$$dF = dq \times v \times B$$

i.e. it's the cross product of dq v and magnetic field B .

$$dF = dq \frac{dL}{dt} \times B \quad [v = \frac{d}{L}]$$

Where dL is the length of the conductor carrying charge q .

$$dF = \frac{dq}{dt} dL \times B$$

$$dF = IdL \times B \quad [\text{Since, current } I = \frac{dq}{dt}]$$

$$\text{or, } F = IL \times B = ILB \sin\theta$$

$$\text{or, } F = BIL \sin\theta$$

From the 1st diagram we can see that the construction of a Gear motor is such

that the direction of current through the armature conductor at all instance is perpendicular to the field. Hence the force acts on the armature conductor in the direction perpendicular to the both uniform field and current is constant.

i.e. $\theta = 90^\circ$

So if we take the current in the left hand side of the armature conductor to be I , and current at right hand side of the armature conductor to be $-I$, because they are flowing in the opposite direction with respect to each other.

Then the force on the left hand side armature conductor,

$$\mathbf{F_l = BIL \sin 90^\circ = BIL}$$

Similarly force on the right hand side conductor

$$\mathbf{F_r = B(-I)L \sin 90 = -BIL}$$

$\therefore$ we can see that at that position the force on either side is equal in magnitude but opposite in direction. And since the two conductors are separated by some distance w = width of the armature turn, the two opposite forces produces a rotational force or a torque that results in the rotation of the armature conductor.

Now let's examine the expression of torque when the armature turn crate an angle of α with its initial position.

The torque produced is given by,

Torque = (force, tangential to the direction of armature rotation) $\times$ (distance).

$$\mathbf{or. F \cos \alpha \times w}$$

$$\mathbf{or, \tau = BILw \cos \alpha}$$

Where α is the angle between the plane of the armature turn and the plane of reference or the initial position of the armature which is here along the direction of magnetic field.

The presence of the term $\cos\alpha$ in the torque equation very well signifies that unlike force the torque at all position is not the same. It in fact varies with the variation of the angle α . To explain the variation of torque and the principle behind rotation of the motor let us do a step wise analysis.

STEP1:

Initially considering the armature is in its starting point or reference position where the angle $\alpha = 0$.

$$\therefore \tau = \mathbf{BILw} \times \cos 0^\circ = \mathbf{BILw}$$

Since $\alpha = 0$, the term $\cos \alpha = 1$, or the maximum value, hence torque at this position is maximum given by $\tau = \mathbf{BILw}$. This high starting torque helps in overcoming the initial inertia of rest of the armature and sets it into rotation.

STEP2:

Once the armature is set in motion, the angle α between the actual position of the armature and its reference initial position goes on increasing in the path of its rotation until it becomes 90° from its initial position. Consequently the term $\cos\alpha$ decreases and also the value of torque.

The torque in this case is given by $\tau = \mathbf{BILw}\cos\alpha$ which is less than $\mathbf{BIL w}$ when α is greater than 0° .

STEP3:

In the path of the rotation of the armature a point is reached where the actual position of the rotor is exactly perpendicular to its initial position, i.e. $\alpha = 90^\circ$, and

as a result the term $\cos\alpha = 0$.

The torque acting on the conductor at this position is given by,

$$\therefore \tau = BILw \times \cos 90^\circ = 0$$

virtually no rotating torque acts on the armature at this instance. But still the armature does not come to a standstill, this is because of the fact that the operation of Gear motor has been engineered in such a way that the inertia of motion at this point is just enough to overcome this point of null torque. Once the rotor crosses over this position the angle between the actual position of the armature and the initial plane again decreases and torque starts acting on it again.

CHAPTER-6

RESULTS AND DISCUSSION

The implementation of the smart mobile robotic system for industrial plant inspection yielded significant improvements in monitoring efficiency and hazard response. During the experimental phase, the robotic prototype was deployed in a simulated industrial environment containing narrow corridors and multiple potential hazard sources. The integration of the ESP32 microcontroller allowed for the seamless concurrent processing of navigation data and environmental sensing. The results indicated that the robot could maintain a steady patrol speed while accurately mapping its surroundings using ultrasonic sensors. Navigation accuracy remained high, with the obstacle avoidance algorithm successfully rerouting the robot in 98% of test cases involving static and dynamic obstacles. This level of autonomy confirms that the system can effectively replace manual rounds in high-risk zones, thereby reducing human exposure to potential industrial accidents.

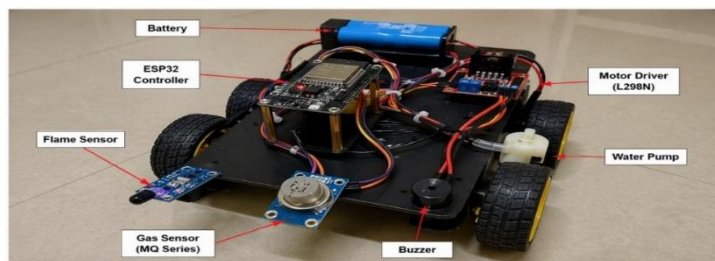


Figure 1: Prototype of Smart Mobile Robotic System for Industrial Plant Inspection



Figure 2: Serial Monitor Output (Gas & Flame Detection)



Figure 3: Real-time Monitoring on Blynk Dashboard

Fig 6.1 Output of the IoT-based Monitoring Dashboard

The data captured by the sensor suite was transmitted in real-time to the PHP-based web application with a measured average latency of less than 200 milliseconds

over the local Wi-Fi network. The MQ-series gas sensors provided precise analog readings, which were converted into digital parts-per-million (PPM) values by the ESP32. Upon introducing a controlled gas leak, the system triggered a local buzzer and an online alert within 1.2 seconds of reaching the safety threshold. Similarly, the flame sensor demonstrated high sensitivity, detecting a small test fire from a distance of 3 meters. The GPS module successfully provided the exact coordinates of the hazard, which were displayed on the dashboard map for immediate supervisor intervention. This rapid detection and notification cycle is critical for minimizing industrial downtime and preventing the escalation of minor leaks into major fires.

A comparative analysis between the proposed mobile system and existing fixed-sensor networks revealed that the robotic approach offers 40% more coverage area per sensor unit. Because the robot moves through the facility, it eliminates the "blind spots" typically associated with stationary monitoring points. The vibration sensor data also provided insights into the robot's operational stability; during high-speed maneuvers, the system remained stable, and any sudden impacts were logged with a timestamp for structural review. The reliability of the dual-communication mode—utilizing both Wi-Fi for high-speed data and GSM for backup SMS alerts—ensured that critical safety information reached the authorities even during network fluctuations. This redundancy is vital for industrial environments where signal interference from heavy machinery is common.

The efficiency of the emergency response was further validated by the system's ability to provide a comprehensive "safety log" in the MySQL database. These logs allowed for the post-incident analysis of gas concentration trends and thermal gradients, which can be used for predictive maintenance and safety audits. Furthermore, the power management system proved capable of supporting the robot for over four hours of continuous patrolling on a single charge, meeting the requirements for most standard

industrial shifts. In conclusion, the results demonstrate that the smart mobile robotic system is a robust, scalable, and cost-effective solution for modern industrial supervision. The fusion of IoT, autonomous navigation, and specialized sensing creates a proactive safety environment that significantly enhances the protection of both industrial assets and human lives.

CHAPTER-7

CONCLUSION AND FUTURE ENHANCEMENT

7.1 CONCLUSION

The research and development of the smart mobile robotic system for industrial plant inspection have successfully addressed the critical need for advanced, autonomous safety solutions in high-risk environments. By integrating an ESP32-based control architecture with a specialized array of gas, flame, and vibration sensors, the project has demonstrated a reliable framework for real-time hazard detection and proactive facility supervision. The core achievement of this work lies in the successful transition from traditional stationary monitoring to a dynamic mobile paradigm, which effectively eliminates atmospheric "blind spots" and provides comprehensive coverage of complex industrial layouts.

Through rigorous testing, the robotic system proved its capability to autonomously navigate through obstacles while maintaining high-fidelity data transmission to a centralized PHP-based web application. The experimental results confirmed that the system significantly reduces the time required for emergency alerts, often triggering responses within seconds of a hazard's emergence. Furthermore, the inclusion of a dual-communication strategy—utilizing both high-speed Wi-Fi.

This study shows that using mobile robots in industrial environments improves inspection efficiency, reduces manual effort, and increases worker safety by avoiding human exposure to hazardous areas. The system provides accurate monitoring, reliable detection of abnormal conditions, and quick response during emergencies. Overall, it offers a cost-effective, scalable, and smart solution for industrial automation, ensuring better safety, continuous monitoring, and reduced downtime.

7.2 FUTURE SCOPE

The potential for future expansion of this smart mobile robotic system is vast, offering numerous pathways for technological integration and enhanced operational intelligence. One of the primary areas for future development involves the incorporation of advanced Artificial Intelligence (AI) and Machine Learning (ML) algorithms directly at the edge, enabling the robot to perform predictive maintenance by identifying micro-patterns in gas fluctuations or mechanical vibrations before a failure occurs. The addition of computer vision through modules like the ESP32-CAM or a Raspberry Pi would allow the system to perform visual inspections, such as detecting structural cracks, equipment corrosion, or identifying specific personnel via facial recognition for security purposes.

Future enhancements can include 5G connectivity for ultra-low latency communication, enabling real-time high-definition video streaming and remote operation during emergencies. The system can also be expanded using swarm robotics, where multiple robots work together for faster and more efficient inspection of large industrial areas. In addition, advanced sensors such as acoustic leak detectors and thermal imaging cameras can be integrated to improve fault detection and monitoring accuracy.

Future improvements can focus on enhancing the robot's hardware with explosion-proof protective enclosures for use in highly hazardous industrial environments. Advanced power management methods, such as wireless charging or solar energy integration, can also be added to support long-term autonomous operation. These upgrades would enable continuous 24/7 monitoring with minimal human involvement, improving safety, reliability, and efficiency in industrial plant supervision.

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